Supplementary Information of "Estimating quantile treatment effect on the original scale of the outcome variable: a case study of common cold treatments"

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2023-10-20

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1. Interpretation of BQTE

Back-transformed quantile treatment effect (BQTE) is a function that describes the difference between the distributions of outcome values of two populations. We will call the two populations that we compare as the **treatment** population and the **control** population, motivated by the need to compare how a particular treatment affects the outcome value in a randomized controlled trial (RCT). In an RCT, the study sample is randomly split into the treatment group (that represents the treatment population) and the control group (that represents the control population). The treatment of interest is applied to the treatment group and placebo is applied to the control group. The possible difference in the outcome distributions between the treatment group and the control group is interpreted as the population-level treatment effect. We will use the BQTE function to quantify this difference between the two populations.

For any possible outcome value x, the theoretical value of BQTE(x) is defined as follows:

1. Find which proportion of the control population has outcome values $\leq x$. Call this proportion $F(x)$.

2. Find what is the corresponding outcome value x^* in the treatment population for which the proportion of the treatment population having outcome values $\leq x^*$ is the same $F(x)$.

3. Define $\text{BQTE}(x) = x^* - x$.

Thus, BQTE describes the difference between the outcome value of the treatment population and the control population when this comparison is done between treated and control individuals whose outcome values rank similarly among their respective populations. As the value of BQTE may vary across outcome values, we present BQTE as a function of outcome value x in the control group.

Mathematically,

$$\text{BQTE}(x) = {F^*}^{-1}\left(F(x)\right) - x,$$

where F and F^* are the cumulative distribution functions of the outcome values of the control and treatment populations, respectively, and ${F^*}^{-1}$ is the inverse function of F^*.

The difference in the means of the populations, called the **average treatment effect** (ATE), is the most common approach to assess the effect of the treatment at the population level. However, the BQTE function measures the difference between the two populations in a more detailed way than the single value of ATE. For example, if we observe that BQTE is not constant across the outcome values, then we can conclude that the individual-level treatment effect (see below for definition) cannot be constant across individuals. Or, if the relative BQTE (defined as $\text{BQTE}(x)/x$) is not constant, then we know that the individual-level treatment effect cannot be proportional to the untreated outcome value by a single constant. Neither of these pieces of information can be inferred from the value of the ATE alone. Next we consider what additional information we can extract from the BQTE function.

1.1. When does BQTE estimate an individual-level treatment effect?

We define two instances of the concept of **treatment effect**. A **population-level treatment effect** is the difference between the outcome distributions of the treatment and control populations and it can be estimated from RCT data by different approaches such as by ATE or BQTE which contain different levels of information about the population-level treatment effect. Instead, an **individual-level treatment effect** is a theoretical construction that cannot, in general, be estimated from an RCT without some additional assumptions. To formally define the individual-level treatment effect, we follow the framework of the potential outcomes by Rubin (1974; Journal of Educational Psychology, 66(5), 688–701). Suppose that we have not yet carried out the experiment. For any individual i, we can imagine two potential outcome variables:

- X_i would be the outcome variable if individual i belonged to the control population,
- X_i^* would be the outcome variable if individual i belonged to the treatment population.

Before the experiment, we consider both X_i and X_i^* as random variables and their difference, $T_i = X_i^* - X_i$, defines a random variable that represents the theoretical treatment effect for individual i. Once we do the experiment, we will observe a value for either X_i (if i was assigned to the control group) or X_i^* (if i was assigned to the treatment group) but never for both X_i and X_i^* simultaneously. Thus, the experiment does not provide a direct way to estimate the treatment effect T_i of individual i. Note that from now on, we will refer by the random variables X and X^* to the untreated and treated potential outcomes of the *same* individual even when we do not specify that same individual by using a common subscript as above.

To conceptually connect BQTE, that quantifies the population-level treatment effect, to the individual-level treatment effects, we will consider the concept of order-preserving treatments.

Definition. We say that a **treatment preserves order** (or that the **treatment is order-preserving**) if the untreated outcome values of a set of individuals have, on average, the same relative order as the treated outcome values of the same set of individuals would have had, if these individuals had been treated. Formally, we define that the treatment preserves order when, for every outcome value x, $F^*\left(\text{E}(X^* \mid X = x)\right) = F(x)$.

For example, consider a trial where the outcome is the duration of the common cold, the treatment group is treated with zinc acetate lozenges and the control group is given placebo. The assumption that the treatment preserves order means that the durations of the colds of the control individuals would have had approximately the same relative order if those individuals had been treated with zinc acetate lozenges rather than given placebo.

While, based on RCT data, we cannot directly confirm that a treatment preserves order, we can make the following remarks.

- First, the assumption that the treatment preserves order seems more plausible the more homogeneous the individuals in the study population are, because within a homogeneous population there are less differences between the individuals that could affect how the treatment applies to each individual. Thus, for example, the assumption of order-preserving treatment seems more plausible in an RCT that studies females between 40-45 years of age than in another RCT that studies both males and females between 18-80 years of age.
- Second, in case that some covariate information were available on the individuals included in the RCT (such as sex, age, medication, disease history etc.), then we would expect that under an order-preserving treatment the way how the covariates predict the outcome values would be similar in the treatment population as it is in the control population. For example, if male individuals tend to have longer disease duration than female individuals in the control group, then the same should be true in the treatment group if the treatment preserves order. If, on the other hand, we observe that the covariates such as sex do not predict the disease duration consistently in both the control and the treatment populations, then the assumption of order-preserving treatment may not be appropriate. Note that the treatment could preserve order in a subpopulation (such as in males only) even if it did not preserve order in a larger population (such as in both sexes together).

The connection between the order-preserving treatments and the individual-level treatment effects can be formulated as follows: **BQTE can be interpreted as the expected value of the individual-level treatment effect if and only if the treatment preserves order.**

To justify the claim, consider a treatment that preserves order, that is, $F^*\left(\mathrm{E}(X^* \mid X = x)\right) = F(x)$ for all x. By applying ${F^*}^{-1}$ on both sides of the equation we get

$$\mathrm{E}(X^* \mid X = x) = {F^*}^{-1}\left(F(x)\right)$$

and since by definition $\mathrm{BQTE}(x) = {F^*}^{-1}(F(x)) - x$ it follows that

$$\mathrm{BQTE}(x) = \mathrm{E}(X^* \mid X = x) - x = \mathrm{E}(X^* - X \mid X = x).$$

Thus, BQTE is the conditional expectation of the individual-level treatment effect and we can interpret BQTE as an estimator of the individual-level treatment effect.

On the other hand, suppose that the treatment does not preserve order. Then there exists some outcome value x in the control population whose quantile level $F(x)$ does not remain constant under the treatment: $F^*\left(\mathrm{E}(X^* \mid X = x)\right) \neq F(x)$. By applying ${F^*}^{-1}$ on both sides of this inequality, and following the steps above, we see that

$$\mathrm{BQTE}(x) \neq \mathrm{E}(X^* \mid X = x) - x = \mathrm{E}(X^* - X \mid X = x).$$

Thus, if the treatment does not preserve order, $\mathrm{BQTE}(x)$ does not match with the conditional expectation of the individual-level treatment effect.

Theoretically, BQTE can be interpreted as the expected value of the individual-level treatment effect if and only if the treatment preserves order. Unfortunately, there is no general way to verify whether a given treatment preserves order based on RCT data. However, we can extract useful information about the individual-level treatment effects from the BQTE function even without an assumption about the preservation of order.

1.2 Bounds on the individual-level treatment effect

In this section, we compute upper and lower bounds on the individual-level treatment effect and further extend these bounds to the average treatment effect in the tails of the distribution.

Consider a treatment that does not necessarily preserve order. Let us denote the range of the possible value of the treatment population by (L^*, U^*), where L^* and U^* refer, respectively, to the lower and upper bounds of the treatment population. We use the following estimators for these bounds:

$$\begin{aligned} \widehat{L^*} &= \min\{x^* \text{ of the treatment group}\}, \\ \widehat{U^*} &= \max\{x^* \text{ of the treatment group}\}, \end{aligned}$$

that is, we use the observed minimum and maximum of the outcome values of the treatment group as the estimates of L^* and U^*, respectively. This leads to an estimated lower bound of $\widehat{L^*} - x$ and an estimated upper bound of $\widehat{U^*} - x$ for the treatment effect of an untreated individual with an outcome value $X = x$. The logic is that since we expect that (almost) all outcome values in the treatment population are within the interval $\left[\widehat{L^*}, \widehat{U^*}\right]$, we expect that the treatment effect at $X = x$ is (almost always) bounded within $\left[\widehat{L^*} - x, \widehat{U^*} - x\right]$. The accuracy of these bounds increases with the sample size of the treatment group and decreases with increasing kurtosis (fatness of the tails) of the outcome distribution of the treatment population.

We can extend the idea of the bounds to the average treatment effect of the upper or lower tail as follows. For the outcome value x, define the **lower tail average treatment effect (LTATE)** as

$$\text{LTATE}(x) = \text{E}\left(X^* \mid X \leq x\right) - \text{E}\left(X \mid X \leq x\right).$$

and the **upper tail average treatment effect (UTATE)** as

$$\text{UTATE}(x) = \text{E}\left(X^* \mid X \geq x\right) - \text{E}\left(X \mid X \geq x\right).$$

While we cannot directly estimate these quantities for a general treatment, we can get a lower bound for LTATE by **lower tail back-transformed quantile treatment effect (LTBQTE)** and an upper bound for UTATE by **upper tail back-transformed quantile treatment effect (UTBQTE)**:

$$\begin{aligned} \text{LTBQTE}(x) &= \text{E}\left(X^* \mid X^* \leq {F^*}^{-1}(F(x))\right) - \text{E}\left(X \mid X \leq x\right) \\ \text{UTBQTE}(x) &= \text{E}\left(X^* \mid X^* \geq {F^*}^{-1}(F(x))\right) - \text{E}\left(X \mid X \geq x\right). \end{aligned}$$

Thus, LTBQTE (UTBQTE) is the difference in the averages of the treatment and control populations when the average is computed for the same proportion of the populations from the lower (upper) tail. Note that LTBQTE(x) approaches ATE as x gets large while UTBQTE(x) approaches ATE as x gets small. Simple estimators to estimate LTBQTE(x) and UTBQTE(x) result by replacing the expectations by the numerical averages.

For these quantities, the following relations apply:

$$\text{LTATE}(x) \geq \text{LTBQTE}(x) \quad \text{and} \quad \text{UTATE}(x) \leq \text{UTBQTE}(x).$$

To verify the argument, let us consider the first inequality. We will show that LTATE(x) $\geq$ LTBQTE(x), which equals to showing that

$$\text{E}\left(X^* \mid X \leq x\right) \geq \text{E}\left(X^* \mid X^* \leq {F^*}^{-1}(F(x))\right).$$

Both quantities in the above inequality are integrals in the marginal space of random variable X^*. The left-hand side is over the event $A = \{X \leq x\}$ and the right-hand side is over the event $A^* = \{X^* \leq {F^*}^{-1}(F(x))\}$.

Let μ be the underlying probability measure of the two-dimensional random variable (X, X^*). By definition of the marginal CDFs F of X and F^* of X^*, both events have the same probability $\mu(A) = \mu(A^*) = F(x)$. Intuitively, among all events with the same probability, the event A^* is the one with the smallest expected value of X^*. To formally prove this, note that for all $\omega \in A \backslash A^*$, $X^*(\omega) > F^{*-1}(F(x))$ and that $\mu(A \backslash A^*) = \mu(A^* \backslash A)$. Hence,

$$
\begin{aligned}
\mathrm{E}\left(X^* \mid X \leq x\right) &= \int_A x^* d\mu(x^*) \\
&= \int_{A \cap A^*} x^* d\mu(x^*) + \int_{A \backslash A^*} x^* d\mu(x^*) \\
&\geq \int_{A \cap A^*} x^* d\mu(x^*) + \mu(A \backslash A^*) \cdot F^{*-1}(F(x)) \\
&= \int_{A \cap A^*} x^* d\mu(x^*) + \mu(A^* \backslash A) \cdot F^{*-1}(F(x)) \\
&\geq \int_{A \cap A^*} x^* d\mu(x^*) + \int_{A^* \backslash A} x^* d\mu(x^*) \\
&= \int_{A^*} x^* d\mu(x^*) \\
&= \mathrm{E}\left(X^* \mid X^* \leq F^{*-1}(F(x))\right).
\end{aligned}
$$

As a conclusion, we have shown that LTBQTE cannot overestimate the average treatment effect in the lower tail.

The condition $\mathrm{UTATE}(x) \leq \mathrm{UTBQTE}(x)$ can be verified similarly.

1.3. Variance of the individual-level treatment effect

In this section, we consider how much the individual-level treatment effect varies across the individuals and how that variance depends on the assumption of order-preserving treatment. For that, we write the variance of the individual-level treatment effect $X^* - X$ in terms of correlation $r = \mathrm{cor}(X^*, X)$ and variance $v = \mathrm{var}(X)$ of the untreated population and $v^* = \mathrm{var}(X^*)$ of the treated population as follows

$$
\begin{aligned}
\mathrm{var}(X^* - X) &= v + v^* - 2\,\mathrm{cov}(X^*, X) \\
&= v + v^* - 2\,r\,\sqrt{v \cdot v^*}.
\end{aligned}
$$

Thus, the variance of the treatment effect is the smallest possible when $r = 1$, in which case the treatment preserves order, and is the largest possible when $r = -1$, in which case the treatment completely flips the order around.

We conclude that the order-preserving treatment with a linear relationship between the treated and untreated outcome values is the most conservative example of treatment in the sense of minimizing the variance of the treatment effect in the population.

2. Validating the `bqte` function

Let's test `bqte()` function by generating data for the treatment (T) group and the control (C) group.

Continuous case. For a given choice of continuous distributions, we can compute the exact BQTE for each possible outcome value in controls by first using cumulative density function (CDF) in controls to get a correct tail probability in the control distribution and then applying the quantile function (inverse of CDF) in the treatment group to get the corresponding outcome value in that group. Difference between the two outcome values is the exact BQTE.

Discrete case. For discrete distributions, the simulation setting is slightly more complex. There, each possible outcome value in controls corresponds to a probability mass which in turn corresponds to a probability interval in CDF. When this interval is mapped by the inverse of CDF in the treatment group, the result contains possibly multiple outcome values in the treatment group. To get the exact BQTE, we compute the expectation of the outcome value in the treatment group conditional on the event that corresponds to the particular interval of CDF. The difference between this expectation and the outcome value in the controls is then the exact BQTE.

We apply three estimators of BQTE. The first two are versions of the estimator presented in our main text and the third was defined by Doksum (1974).

1. **Bagging estimator** that equals to the average of the BQTE estimates across the bootstrap samples. This is our default estimator and it is shown in Figures unless otherwise stated in the Figure caption. The motivation for bootstrap averaging ("bagging") is that it can reduce variance of the estimator in some contexts, see Fig. S4.

2. **Direct estimator** where we apply our BQTE estimator to the original data and not to the bootstrapped samples as in bagging. (In `bqte()`, this estimator can be chosen by setting `bagging = FALSE`.)

3. **Doksum estimator** defined as $\widehat{\Delta}(x) = y_{i(x)} - x$ where index $i(x) = \left\lfloor n\,\widehat{F}(x) \right\rfloor + 1$ with $\widehat{F}$ being the empirical cumulative distribution function of the control group, n the sample size of the treatment group and y_i the value in the treatment group having rank i.

2.1 Generating continuous example data

To study the continuous case, let's assume

$$T \sim \text{Weibull}(1.2, 2) \text{ and } C \sim \text{Weibull}(3, 5).$$

Figure S1 demonstrates this setting and shows the estimates of treatment effect from `bqte()` function.

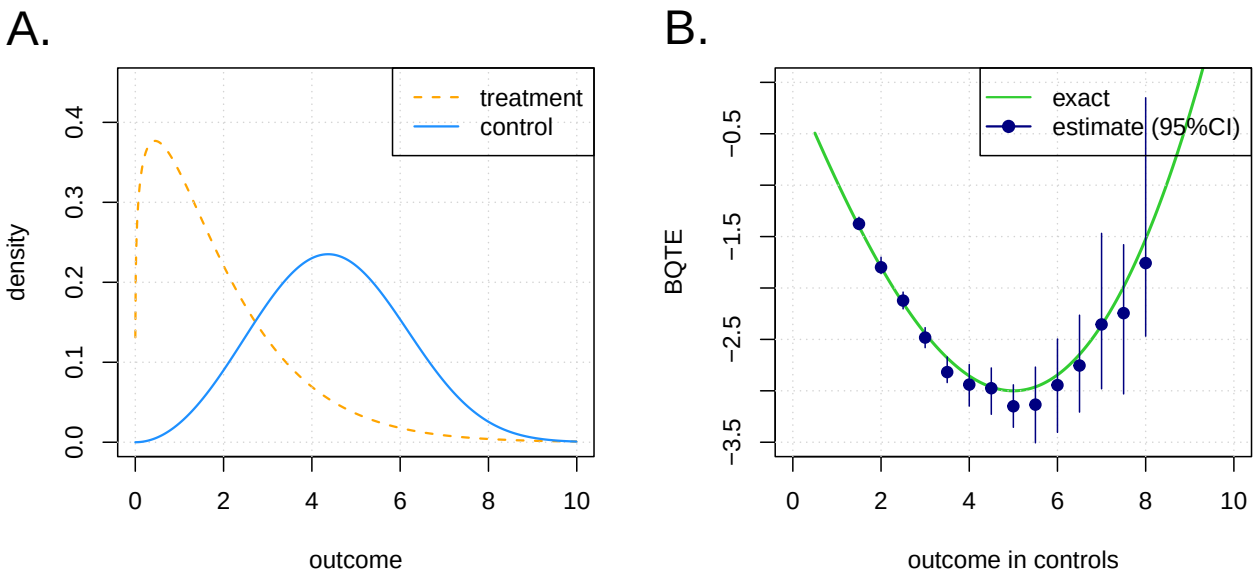

Figure S1. Panel A shows the theoretical distributions in the two groups: Weibull$(1.2, 2)$ in the treatment group and Weibull$(3, 5)$ in the control group. Panel B shows the exact back-transformed quantile treatment effect (BQTE) as a function of the outcome value in controls (green line), together with estimates and their 95% CIs when `bqte()` was applied on a simulated data set with $N = 500$ individuals in each group.

2.2. The range of valid estimates

By default, `bqte()` estimates BQTE in the interval between the empirical quantile levels of $\frac{5}{N}$ and $\frac{N-5}{N}$, where N is the sample size in the control group. For example, in Fig. S1B, $N = 500$ and hence the range of estimation is between the 1st and the 99th percentile of the observed data, which correspond to the values of 2 and 8, respectively, when rounded to integers.

Let's confirm that the default range for estimation leads to well calibrated confidence intervals (CIs), and let's also check that the estimates at more extreme quantile levels of $\frac{1}{N}$ and $\frac{N-1}{N}$ would not anymore remain correctly calibrated.

For both values of $N = 50$ and $N = 500$, we generate $R = 100$ data sets as above from the two Weibull distributions and apply `bqte()` function to estimate BQTE at the outcome values that correspond to the following 5 quantile levels in controls: $\frac{1}{N}, \frac{5}{N}, \frac{1}{2}, \frac{N-5}{N}, \frac{N-1}{N}$. For each N and each quantile level, we then compute the empirical coverage of the 95%CIs and the empirical mean of the BQTE estimates over the R data sets. Results are show in Fig S2.

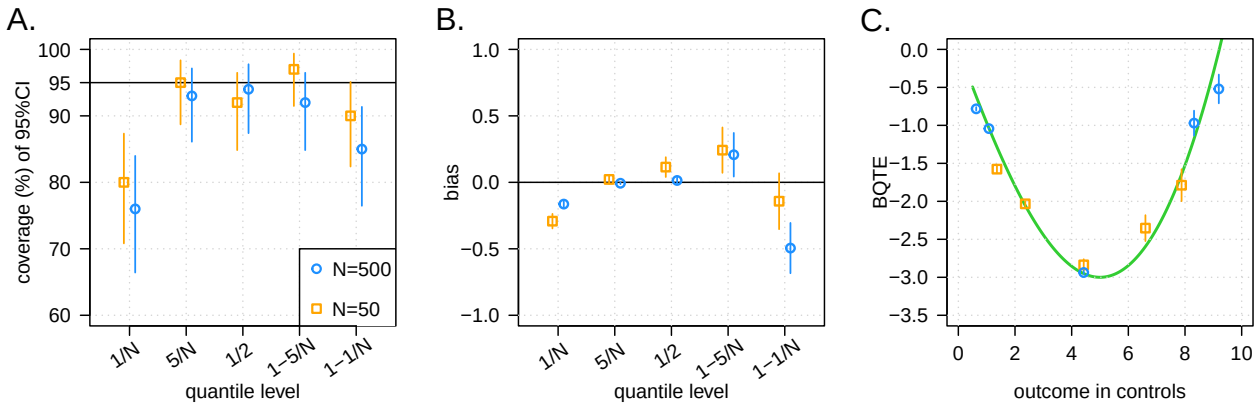

Figure S2. Observed coverage of 95% confidence intervals (panel A), observed bias (panel B) and the mean of the estimates over the 100 replicates compared to the exact values shown by the green line (panel C). The value of N is shown in the legend of panel A. The intervals around each point show 95% confidence intervals based on the sample size of 100 replicates.

In Fig. S2A we see that, independent of sample size N, the 95%CIs are well calibrated for quantile levels $\frac{5}{N}$ and $\frac{N-5}{N}$ but not for more extreme levels of $\frac{1}{N}$ or $\frac{N-1}{N}$. Fig. S2B shows that there is more bias at the extreme quantile levels than near the median for both sample sizes and the bias does not depend much on the sample size except at the extreme quantile levels. (Note, however, that for any fixed outcome value in controls, the bias would disappear as sample size grows; this is demonstrated by the estimates at the median point in panel B where the bias disappears as N grows from 50 to 500.) From Fig. S2C we observe how these biases correspond to some overestimation of the magnitude of the quantile treatment effect (QTE) at the extreme quantile levels of $\frac{1}{N}$ and $\frac{N-1}{N}$. Based on these results, our default option is to restrict the estimation of BQTE and QTE between quantile levels of $\frac{5}{N}$ and $\frac{N-5}{N}$.

Figure S2 shows only the bagging estimator but we also evaluated the other two estimators on the same simulation setting. The root mean square error (RMSE) of the three estimators at the quantile levels of $5/N$, $1/2$ and $1-5/N$ were, for the sample size of 50:

```
##   quantile       at RMSE_bagging RMSE_direct RMSE_doksum
## 1      5/N 2.361544    0.1675442   0.1631935   0.1622676
## 2      1/2 4.424985    0.3955057   0.3958338   0.4191274
## 3    1-5/N 6.602502    0.9003744   0.9023805   1.1436505
```

and for the sample size of 500:

```
##   quantile       at RMSE_bagging RMSE_direct RMSE_doksum
## 1      5/N 1.079022   0.03678602  0.02764578  0.02488011
## 2      1/2 4.424985   0.11279004  0.11264762  0.11370297
## 3    1-5/N 8.318632   0.85930903  0.85459130  0.96972486
```

2.3. The number of quantile points used in estimation

The `bqte()` function takes as input the number K of quantile points, corresponding to the probability values of $1/(K+1), 2/(K+1), \ldots, K/(K+1)$, that are used in the estimation of BQTEs. Between these points, BQTE is estimated by linear interpolation. By default, we set $K = N$, where N is the size of the control group. The value N is a natural upper bound for K because when $K = N$, there is already a separate estimate corresponding to each control observation. But could some smaller value for $K < N$ be a better choice? Let's use a similar simulation procedure as in Fig. S2 to assess how smaller values of $K < N$ compare to our default choice $K = N$.

For this simulation, the sample size in both groups takes values $N = 50$ or $N = 500$, and for both values of N, we will run 100 simulations for each value of $K = \frac{N}{4}, \frac{N}{2}, \frac{2N}{3}, N$. The results are in Fig. S3.

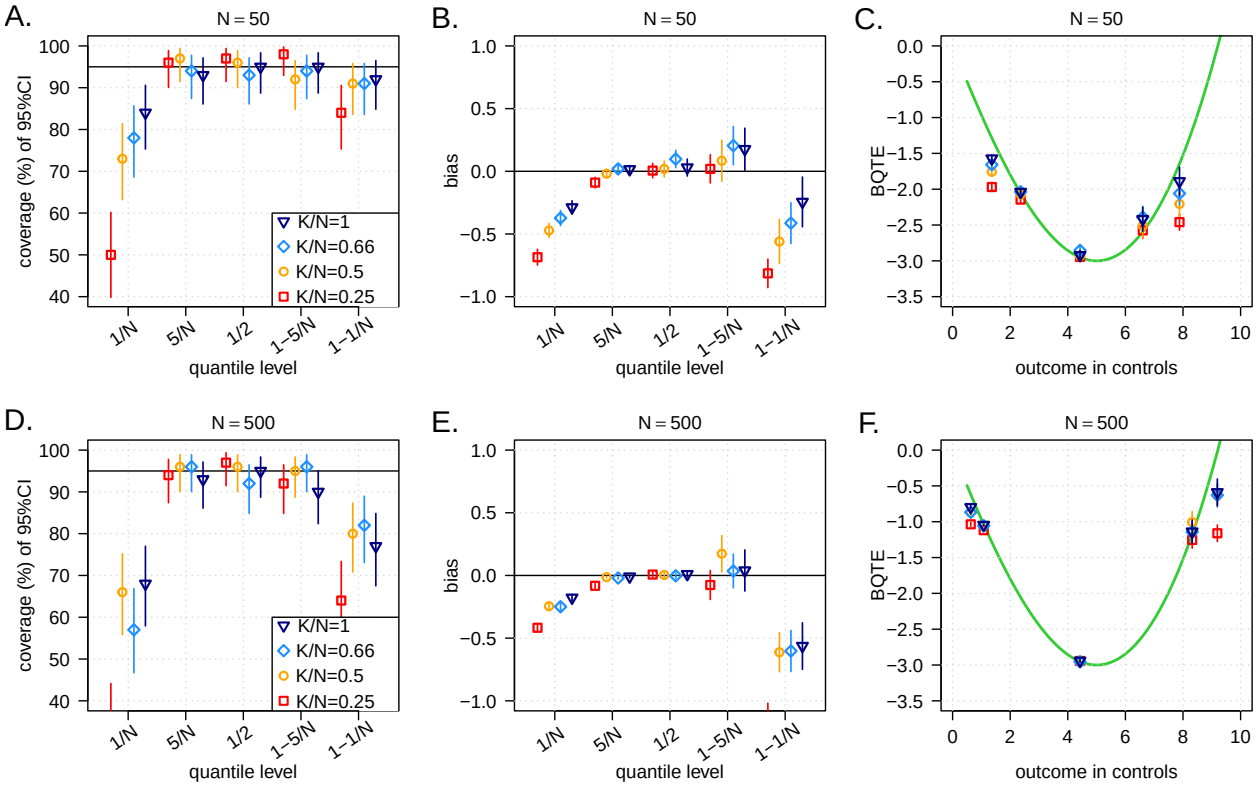

Figure S3. Observed coverage of 95% confidence intervals (panels A, D), observed bias (panels B, E) and the mean of the estimates over the 100 replicates compared to the exact values shown by the green line (panels C, F). The ratio K/N is shown in the legend of panels A and D, and N is shown in the title of each panel. The intervals around each point show 95% confidence intervals based on the sample size of 100 replicates.

In Figure S3, we do not observe clear difference between different values of K/N except at the extreme quantile levels of $1/N$ and $1 - 1/N$, and hence the choice $K = N$ seems a suitable default value to be used in `bqte()` function.

2.4. Magnitude of estimation error and coverage of confidence intervals in discrete data

We used the Mossad data from the main paper to generate realistic data about the duration of cold in a randomized trial when the duration is measured in days. We focus on the recommended interval between quantile levels $5/N$ and $1 - 5/N$ that in these data corresponds to interval (3,17). For each integer value in this interval, we computed the exact BQTE values for the observed distribution of the Mossad data using function `exact.bqte.discrete()`. Then we generated 100 simulations from the empirical distribution from Mossad data with sample sizes equaling to the Mossad data (Treatment group = 49, Control group = 50). We estimated BQTE by all three estimators. Figure S4 shows the root mean squared error of the three estimators as well as the coverage of the 95% confidence interval by our bootstrap method.

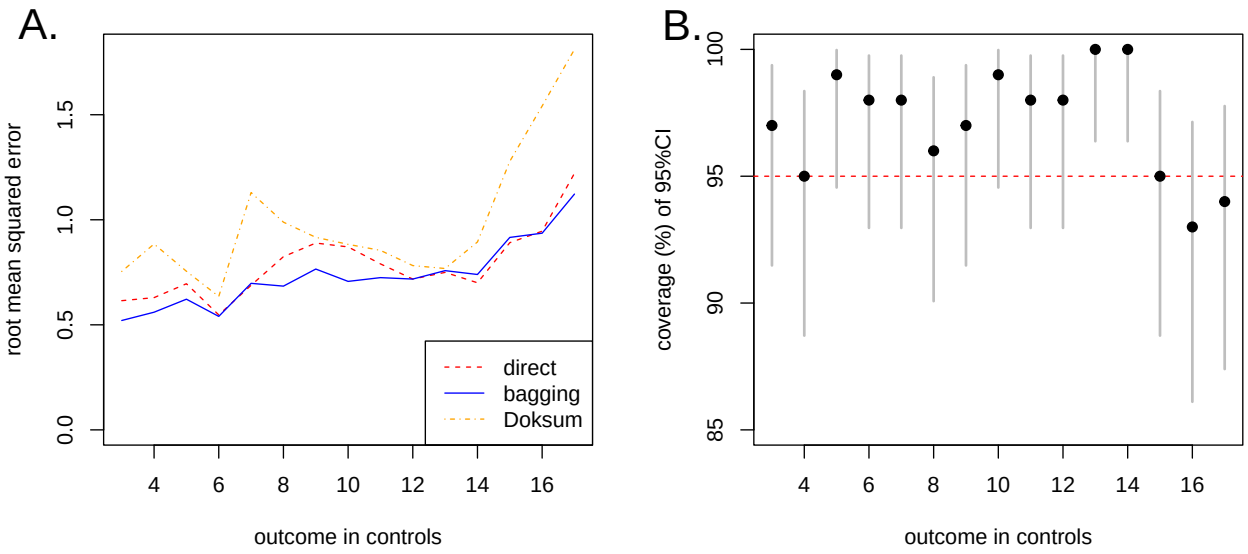

Figure S4. Panel A shows root mean squared error (RMSE) for three estimators (bagging in blue, direct in red and Doksum in orange) at each of the integer valued cold duration between 3 and 17 days. The RMSE is estimated over 100 simulated data sets that follow the observed distributions of the Doksum data from the main text. Panel B shows the empirical coverage of the 95% confidence intervals estimated with bootstrap at the same grid of values as in panel A. The intervals around the point estimates are 95% binomial confidence intervals based on 100 samples and the red line denotes the target coverage of 95%.

From Figure S4 we see that the RMSE seems the smallest for the bagging estimator. As in the continuous setting, there were little differences between the three estimators, we assume that the differences seen here are due to the discreteness of the data. Based on the result of Figure S4A, we have used the bagging estimator as our default estimator in the discrete data sets that we have analyzed in the main text.

Panel B shows that the 95% confidence intervals do not show any evidence of falling short of the target coverage of 95% as the confidence intervals of the coverage estimates in panel B either overlap with the target value or remain above the target value in every case.

3. Source of data of the included examples

Mossad study. The data are from pages 2-4 of the Additional file 2 of https://doi.org/10.1186/s12874-017-0356-y.

Zinc acetate studies. The data are from page 8 of the Supplementary file 2 of https://doi.org/10.1093/ofid/ofx059

Carrageenan studies. The data are from page 3 of the Supplementary file of https://doi.org/10.1002/prp2.810

4. Supplementary figures of real data sets

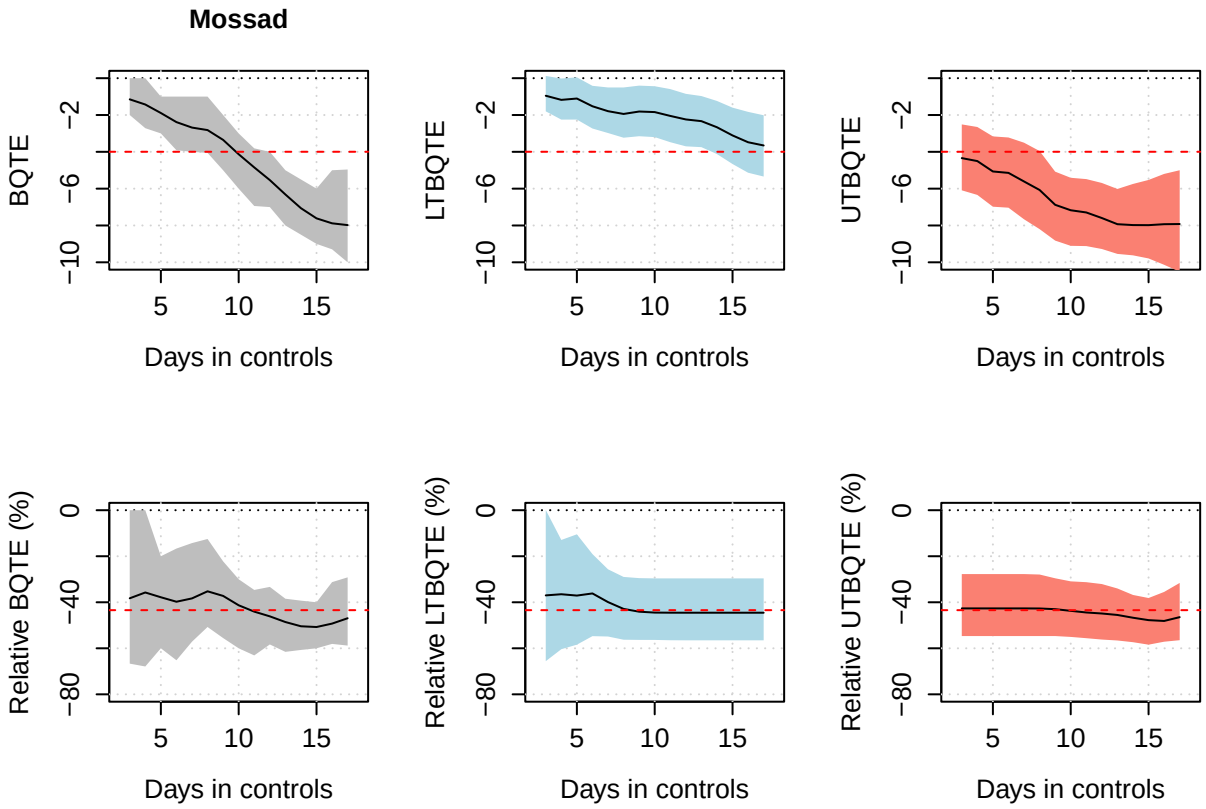

Figure S5. BQTE, LTBQTE and UTBQTE functions for the Mossad data set. The top row shows the values on the outcome scale. The bottom row shows the values on the relative scale. The solid black lines shows the estimates and the band shows the 95% confidence interval based on the bootstrap samples. The horizontal dashed red lines mark the average treatment effect.

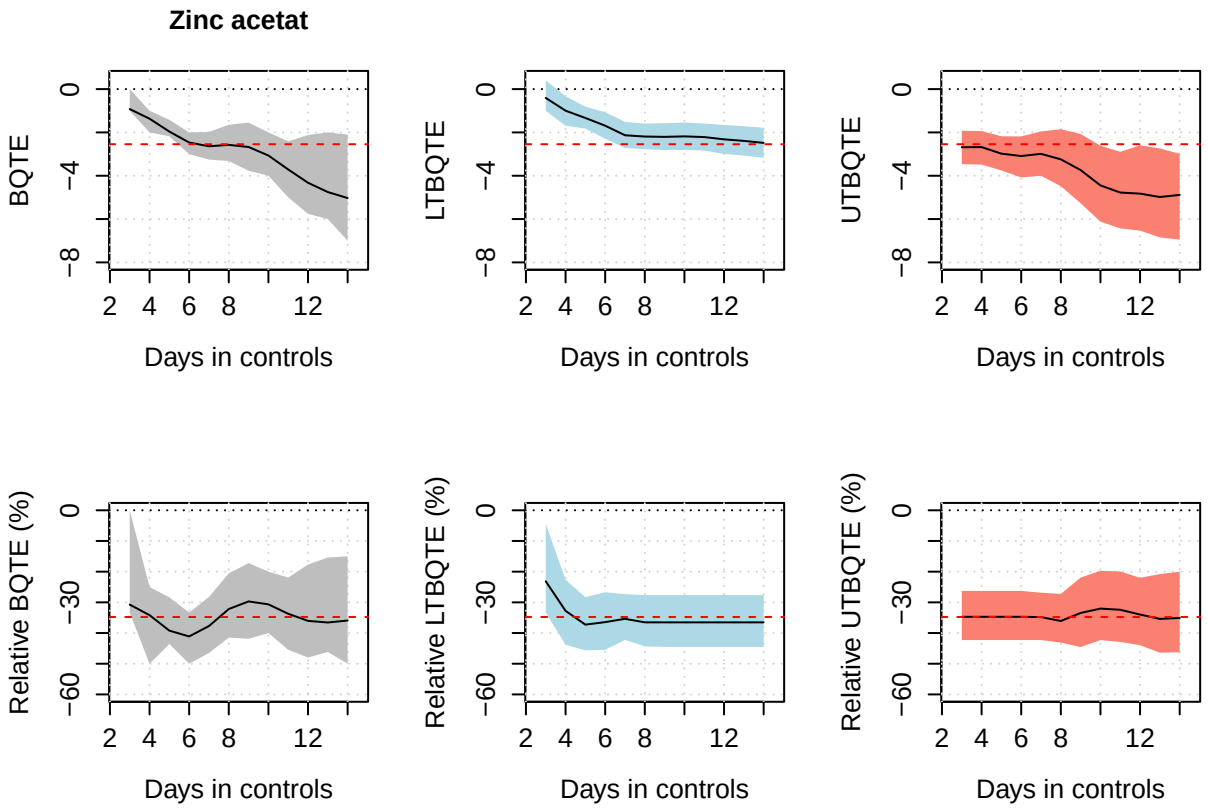

Figure S6. BQTE, LTBQTE and UTBQTE functions for the Zinc acetat data set. The top row shows the values on the outcome scale. The bottom row shows the values on the relative scale. The solid black lines shows the estimates and the band shows the 95% confidence interval based on the bootstrap samples. The horizontal dashed red lines mark the average treatment effect.

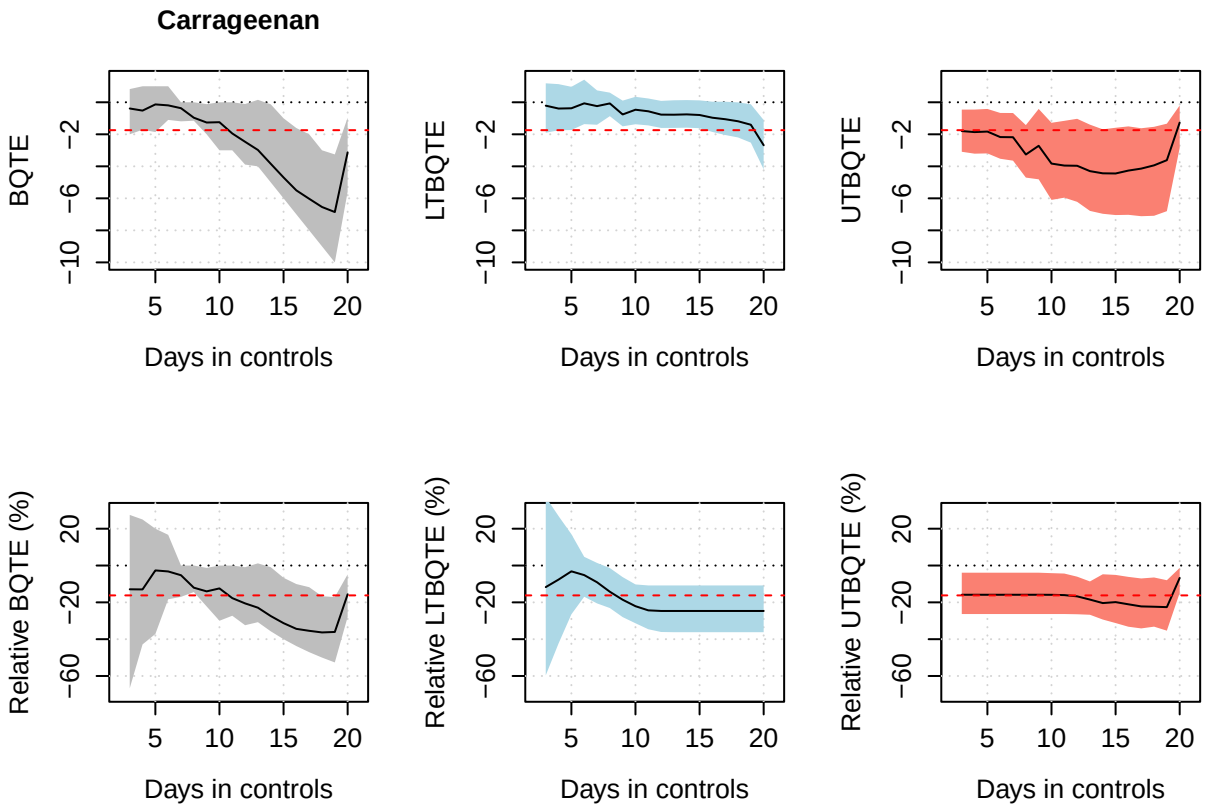

Figure S7. BQTE, LTBQTE and UTBQTE functions for the Carrageenan data set. The top row shows the values on the outcome scale. The bottom row shows the values on the relative scale. The solid black lines shows the estimates and the band shows the 95% confidence interval based on the bootstrap samples. The horizontal dashed red lines mark the average treatment effect.

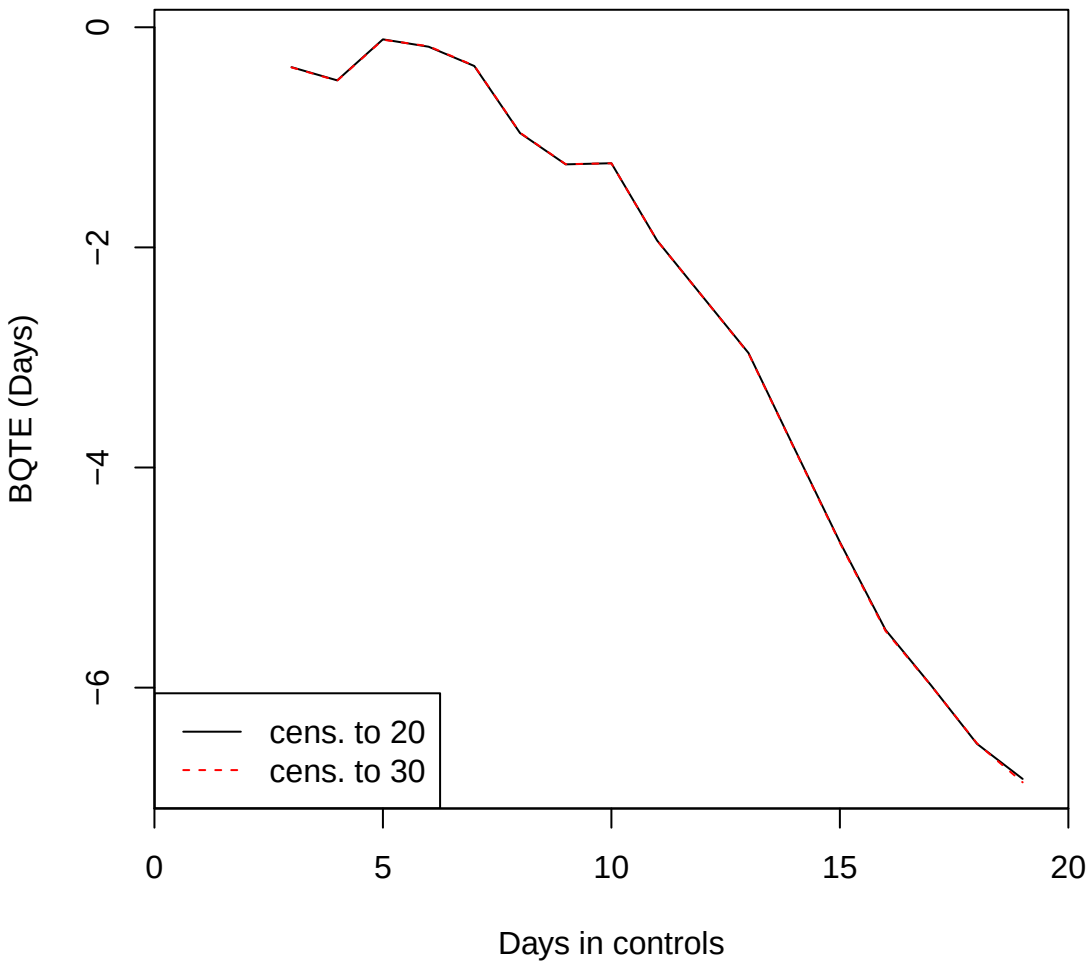

Figure S8. Effect of imputation of the censored observations in the Carrageenan data set. The continuous black line shows the BQTE function when the censored observations were imputed using the value of 20 days. The dashed red line shows the BQTE function when the censored observations were imputed using the value of 30 days. In this calculation, we used B = 20 000 bootstrap samples to ensure stable estimates. There is only a very small difference between the curves at the right-hand side, indicating that the curves are robust to the type of imputation applied in this case.